

Lecture Notes on Thermodynamic Computing

Chapter 8: Applications and Future Directions

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1 8.1 IoT and Edge Computing Applications

• Ultra-Low Power Sensor Nodes

Energy harvesting compatibility:

$$P_{harvested} \sim \mu\text{W to mW range} \quad (1)$$

Thermodynamic processor power budget:

$$P_{processor} = P_{active} \times \text{Duty cycle} + P_{standby} \quad (2)$$

Typical energy harvesting sources:

- Solar: $10 - 100 \text{ W/cm}^2$ (indoor)
- Thermal: $1 - 10 \text{ W/cm}^2$ (5K gradient)
- Vibration: $1 - 100 \text{ W/cm}^3$
- RF: $0.1 - 1 \text{ W/cm}^2$ (far field)

• Ambient Computing Systems

Environmental data processing:

$$\text{Data} = \{T, P, H, \text{Air quality, Motion}\} + \boldsymbol{\eta}_{sensor} \quad (3)$$

Approximate signal processing:

$$\hat{x}[n] = x[n] + \epsilon_{quantization} + \epsilon_{thermal} \quad (4)$$

Information extraction efficiency:

$$\eta_{info} = \frac{H(\text{useful information})}{H(\text{raw data})} \times \frac{P_{raw}}{P_{processed}} \quad (5)$$

- **Wearable Health Monitoring**

Biomedical signal characteristics:

- ECG: 0.1-100 Hz, 1-5 mV amplitude
- EEG: 0.5-50 Hz, 10-100 V amplitude
- EMG: 10-500 Hz, 0.1-5 mV amplitude

Thermal noise floor for biosignals:

$$V_{noise} = \sqrt{4k_B T R_{electrode} \Delta f} \quad (6)$$

For $R_{electrode} = 10$ k, $BW = 100$ Hz:

$$V_{noise} = 4.1 \text{ V RMS at } 300\text{K} \quad (7)$$

SNR for neural signals:

$$SNR = 20 \log_{10} \left(\frac{V_{signal}}{V_{noise}} \right) = 20 \log_{10} \left(\frac{50 \times 10^{-6}}{4.1 \times 10^{-6}} \right) = 21.7 \text{ dB} \quad (8)$$

2 8.2 Machine Learning and AI Applications

- **Neural Network Accelerators**

Synaptic weight storage in kT-RAM:

$$w_{ij} = \frac{Q_{stored}}{Q_{reference}} + \epsilon_{thermal} \quad (9)$$

Weight precision requirements:

$$\text{Precision} = \frac{\text{Weight range}}{\sigma_{thermal}} > 10 - 20 \quad (10)$$

Energy per MAC operation:

$$E_{MAC} = E_{multiply} + E_{accumulate} \approx 5 - 10 \times k_B T \quad (11)$$

- **Bayesian Deep Learning**

Weight uncertainty representation:

$$w \sim \mathcal{N}(\mu_w, \sigma_w^2) \quad (12)$$

Forward pass with uncertainty propagation:

$$y = f(x, w) = f(x, \mu_w) + \epsilon_{uncertainty} \quad (13)$$

Epistemic uncertainty:

$$\sigma_{epistemic}^2 = \mathbb{E}_w[(f(x, w) - \mathbb{E}_w[f(x, w)])^2] \quad (14)$$

Aleatoric uncertainty (from thermal noise):

$$\sigma_{aleatoric}^2 = \mathbb{E}[\epsilon_{thermal}^2] \quad (15)$$

- **Reinforcement Learning Hardware**

Q-value updates with thermal noise:

$$Q(s, a) \leftarrow Q(s, a) + \alpha[r + \gamma \max_{a'} Q(s', a') - Q(s, a)] + \eta(t) \quad (16)$$

Exploration via thermal fluctuations:

$$\pi(a|s) = \frac{\exp((Q(s, a) + \eta(t))/\tau)}{\sum_{a'} \exp((Q(s, a') + \eta(t))/\tau)} \quad (17)$$

Temperature parameter:

$$\tau = \frac{k_B T_{physical}}{q V_{scale}} \quad (18)$$

3 8.3 Signal Processing Applications

- **Software Defined Radio**

Thermal noise-enhanced ADC:

$$\text{SNDR} = 6.02N + 1.76 + 10 \log_{10} \left(\frac{f_s}{2f_{BW}} \right) - \text{Thermal_penalty} \quad (19)$$

Direct RF sampling:

$$f_{sample} > 2 \times (f_{RF} + f_{BW}) \quad (20)$$

Power consumption scaling:

$$P_{ADC} \propto f_{sample} \times 2^{ENOB} \times \text{Technology_factor} \quad (21)$$

- **Cognitive Radio Systems**

Spectrum sensing with uncertainty:

$$\text{Decision} = \begin{cases} H_0 : \text{Spectrum free} & \text{if } E < \gamma_0 \\ H_1 : \text{Spectrum occupied} & \text{if } E > \gamma_1 \\ \text{Uncertain} & \text{if } \gamma_0 \leq E \leq \gamma_1 \end{cases} \quad (22)$$

Detection probability with thermal noise:

$$P_D = Q \left(\frac{\gamma - (P_s + P_n)}{\sqrt{2P_n(P_s + P_n)/N}} \right) \quad (23)$$

- **Compressed Sensing**

Measurement matrix with thermal noise:

$$\mathbf{y} = \Phi \mathbf{x} + \mathbf{w} \quad (24)$$

Recovery condition (RIP):

$$(1 - \delta_s) \|\mathbf{x}\|_2^2 \leq \|\Phi \mathbf{x}\|_2^2 \leq (1 + \delta_s) \|\mathbf{x}\|_2^2 \quad (25)$$

Thermal noise impact on recovery:

$$\|\hat{\mathbf{x}} - \mathbf{x}\|_2 \leq C_1 \sigma_w + C_2 \frac{\sigma_s(\mathbf{x})_k}{\sqrt{k}} \quad (26)$$

4 8.4 Scientific Computing Applications

• Monte Carlo Simulations

Natural random number generation:

$$X \sim \mathcal{N}(0, 1) \text{ from thermal noise} \quad (27)$$

Transformation for arbitrary distributions:

$$Y = F^{-1}(\Phi(X)) \quad (28)$$

Convergence rate:

$$\text{Error} \propto \frac{1}{\sqrt{N_{\text{samples}}}} \quad (29)$$

Energy per sample:

$$E_{\text{sample}} = E_{\text{generation}} + E_{\text{transformation}} + E_{\text{computation}} \quad (30)$$

• Molecular Dynamics

Langevin equation integration:

$$m \frac{dv}{dt} = F(r) - \gamma mv + \sqrt{2\gamma mk_B T} \xi(t) \quad (31)$$

Thermal force implementation:

$$F_{\text{thermal}} = \sqrt{\frac{2\gamma mk_B T}{\Delta t}} \xi \quad (32)$$

Energy conservation check:

$$\Delta E_{\text{total}} = \Delta E_{\text{kinetic}} + \Delta E_{\text{potential}} + E_{\text{dissipated}} \quad (33)$$

• Climate Modeling

Stochastic parameterization:

$$\frac{\partial T}{\partial t} = \nabla \cdot (D \nabla T) + S + \eta_{\text{stochastic}} \quad (34)$$

Ensemble forecasting:

$$\langle T \rangle = \frac{1}{N} \sum_{i=1}^N T_i + \epsilon_{\text{sampling}} \quad (35)$$

Uncertainty quantification:

$$\sigma_T^2 = \frac{1}{N-1} \sum_{i=1}^N (T_i - \langle T \rangle)^2 \quad (36)$$

5 8.5 Cryptography and Security

• True Random Number Generation

Entropy extraction from thermal noise:

$$H(\text{output}) = - \sum_i P(x_i) \log_2 P(x_i) \quad (37)$$

von Neumann debiasing:

$$\text{Input: } 01 \rightarrow 0, \quad 10 \rightarrow 1, \quad 00, 11 \rightarrow \text{discard} \quad (38)$$

Min-entropy estimation:

$$H_{\min} = -\log_2(\max_i P(X_i)) \quad (39)$$

• Physical Unclonable Functions

PUF response with thermal variations:

$$R = f(\text{Challenge}, \text{Process variations}, T_{\text{thermal}}) \quad (40)$$

Reliability metric:

$$\text{Reliability} = P(R_1 = R_2 | \text{same challenge, different time}) \quad (41)$$

Uniqueness metric:

$$\text{Uniqueness} = P(R_1 \neq R_2 | \text{same challenge, different chips}) \quad (42)$$

• Approximate Homomorphic Encryption

Noisy ciphertext:

$$c = \text{Encrypt}(m) + \epsilon_{\text{thermal}} \quad (43)$$

Homomorphic operation:

$$c_3 = c_1 \oplus c_2 = \text{Encrypt}(m_1 \oplus m_2) + \epsilon_{\text{accumulated}} \quad (44)$$

Noise growth bound:

$$||\epsilon_{\text{accumulated}}|| \leq B \cdot L \quad (45)$$

where B is noise bound and L is computation depth.

6 8.6 Emerging Application Domains

• Quantum-Classical Interfaces

Thermal decoherence modeling:

$$\rho(t) = \sum_k K_k \rho(0) K_k^\dagger \quad (46)$$

Kraus operators for thermal bath:

$$K_k = \sqrt{p_k} |k\rangle \langle k| \quad (47)$$

Classical error correction for quantum states:

$$|\psi_{\text{corrected}}\rangle = \mathcal{R}(|\psi\rangle + |\epsilon_{\text{thermal}}\rangle) \quad (48)$$

• Neuromorphic Computing

Synaptic plasticity with thermal noise:

$$\frac{dw}{dt} = \eta(x_{pre}x_{post} - w) + \sqrt{2D}\xi(t) \quad (49)$$

Spike timing with jitter:

$$t_{spike} = t_{nominal} + \epsilon_{thermal} \quad (50)$$

Information capacity:

$$C = \log_2 \left(1 + \frac{P_{signal}}{P_{thermal} + P_{shot}} \right) \quad (51)$$

• DNA Computing

Thermal fluctuation effects:

$$P_{hybridization} = \frac{1}{1 + \exp(\Delta G/k_B T)} \quad (52)$$

Error rate in DNA synthesis:

$$\epsilon_{DNA} = \epsilon_{polymerase} + \epsilon_{thermal} \quad (53)$$

Storage density:

$$\rho = \frac{\text{bits}}{\text{volume}} = \frac{2N}{N_A \times M_{DNA}/\rho_{DNA}} \quad (54)$$

7 8.7 Technology Roadmap

• Near-term Developments (2025-2030)

Process technology improvements:

- 3nm FinFET optimization for subthreshold
- Advanced materials (III-V, 2D materials)
- Through-silicon vias for 3D integration

Performance projections:

$$\frac{E_{2030}}{E_{2025}} \approx 0.1 - 0.3 \quad (55)$$

Density scaling:

$$\text{Density}_{2030} \approx 10 \times \text{Density}_{2025} \quad (56)$$

- **Medium-term Goals (2030-2040)**

Novel device concepts:

- Tunneling FETs (TFETs)
- Negative capacitance transistors
- Spin-based devices
- Memristive computing

System integration advances:

$$\text{Integration level} = \frac{\text{Sensors} + \text{Processing} + \text{Storage} + \text{Communication}}{\text{Package volume}} \quad (57)$$

- **Long-term Vision (2040+)**

Fundamental limits approach:

$$E_{operation} \rightarrow k_B T \ln(2) = 2.9 \times 10^{-21} \text{ J at 300K} \quad (58)$$

New computing paradigms:

- Biological-electronic hybrids
- Molecular electronics
- Quantum-thermal interfaces

8.8 Research Challenges and Opportunities

- **Fundamental Research Questions**

Information-energy trade-offs:

$$H(\text{information}) \leq \frac{E_{available}}{k_B T \ln(2)} \quad (59)$$

Quantum thermodynamics:

$$\Delta S \geq \frac{Q_{dissipated}}{T} \quad (60)$$

Computation at the thermodynamic limit:

$$P_{min} = k_B T \frac{dH(X)}{dt} \quad (61)$$

- **Engineering Challenges**

Process variation control:

$$\frac{\sigma_{V_{th}}}{V_{th}} < 0.1 \text{ for reliable operation} \quad (62)$$

Temperature stabilization:

$$\Delta T < \frac{k_B T}{q} \text{ for stable operation} \quad (63)$$

Packaging and thermal management:

$$R_{thermal} \times P_{dissipated} < \Delta T_{allowed} \quad (64)$$

• Interdisciplinary Opportunities

Biology-inspired computing:

- Neural spike processing
- Molecular recognition
- Evolutionary algorithms

Materials science contributions:

- 2D materials (graphene, MoS)
- Topological insulators
- Organic semiconductors

Physics collaborations:

- Non-equilibrium thermodynamics
- Information theory
- Quantum information

9 Examples

Example 9.1 (IoT Sensor Node Energy Budget). Design energy budget for environmental monitoring node.

Given:

- Solar panel: 1 cm², 100 W peak power
- Battery capacity: 100 mAh at 3.3V
- Measurement frequency: once per minute
- Data transmission: once per hour
- Required lifetime: 5 years

Solution:

Daily energy budget:

$$E_{daily} = \frac{100 \times 10^{-3} \times 3.3 \times 3600}{5 \times 365} \quad (65)$$

$$= 653 \text{ J per day} \quad (66)$$

Solar energy available (8 hours sun):

$$E_{solar} = 100 \times 10^{-6} \times 8 \times 3600 = 2.88 \text{ J per day} \quad (67)$$

Energy per measurement cycle:

$$E_{measurement} = \frac{653}{24 \times 60} = 0.45 \text{ J} \quad (68)$$

With thermodynamic processor (1000× improvement):

$$E_{thermo} = \frac{0.45}{1000} = 0.45 \text{ mJ} \quad (69)$$

This enables solar-powered operation!

Example 9.2 (ML Accelerator Comparison). Compare energy efficiency of thermodynamic vs conventional ML accelerator.

Given:

- Neural network: 1M parameters, 1G MAC operations
- Conventional accelerator: 10 pJ/MAC
- Thermodynamic accelerator: 50 kBT/MAC at 300K
- Inference rate: 1000 fps

Solution:

Conventional energy per inference:

$$E_{conv} = 10^9 \times 10 \times 10^{-12} = 10 \text{ mJ} \quad (70)$$

Thermodynamic energy per MAC:

$$E_{MAC,thermo} = 50 \times 1.38 \times 10^{-23} \times 300 \quad (71)$$

$$= 207 \times 10^{-21} \text{ J} = 0.207 \text{ fJ} \quad (72)$$

Thermodynamic energy per inference:

$$E_{thermo} = 10^9 \times 0.207 \times 10^{-15} = 0.207 \text{ J} \quad (73)$$

Power comparison at 1000 fps:

$$P_{conv} = 10 \times 10^{-3} \times 1000 = 10 \text{ W} \quad (74)$$

$$P_{thermo} = 0.207 \times 10^{-6} \times 1000 = 0.207 \text{ mW} \quad (75)$$

Energy efficiency improvement: $\frac{10}{0.207 \times 10^{-3}} = 48,000 \times$

10 Homework Problems

Assignment 8.1: Applications and Future Directions

1. IoT System Design:

- Design complete environmental sensor node
- Calculate energy harvesting requirements

- Optimize duty cycle for 10-year operation
2. **ML Accelerator Architecture:**
 - Design thermodynamic CNN accelerator
 - Implement approximate computing techniques
 - Compare with state-of-the-art conventional designs
 3. **Communication System:**
 - Design cognitive radio using thermodynamic computing
 - Implement spectrum sensing algorithms
 - Optimize power-performance trade-offs
 4. **Cryptographic Applications:**
 - Design thermal-noise-based TRNG
 - Implement PUF using process variations
 - Analyze security properties
 5. **Scientific Computing:**
 - Implement Monte Carlo simulation accelerator
 - Design molecular dynamics co-processor
 - Optimize for specific application domains
 6. **Technology Roadmap:**
 - Project thermodynamic computing evolution
 - Identify key technological barriers
 - Propose research directions
 7. **Interdisciplinary Applications:**
 - Explore bio-inspired thermodynamic computing
 - Design quantum-classical interfaces
 - Investigate novel material applications
 8. **System Integration:**
 - Design heterogeneous computing platform
 - Implement adaptive resource management
 - Optimize for multiple application domains

Course Summary: This completes our comprehensive study of thermodynamic computing, from fundamental principles through device physics to complete system architectures and applications. The field represents a paradigm shift toward computing at the fundamental energy limits, opening new possibilities for ultra-low power systems and novel computational approaches.

Key Takeaways:

- Thermodynamic computing enables $10^3 - 10^6$ energy reductions
- Thermal noise becomes a computational resource
- Probabilistic and approximate computing are essential
- Applications span IoT, AI, communications, and scientific computing
- Significant research opportunities remain in materials, devices, and algorithms